Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-5 — Properties of Logarithms

Project Task — Red Sea Tourism Growth

A 20-minute group project. Work in fours. Every member signs the sheet.

Student name :	Date : .....
Group members :	
MATERIALS This sheet · Squared paper · A pencil and ruler · A calculator	

THE DRIVING QUESTION

A model says visitors to a Red Sea destination grow by the same percentage every year. In which year does the model pass a target — and how does the answer change if the growth rate changes?

THE SITUATION

A classroom model (invented for practice, not a forecast): $20\,000$ visitors in year 1, growing by 20% each year, and each visitor spends a model $1\,500$ SAR.

Planners want the first year with more than $100\,000$ visitors, and the model revenue in that year.

Each group then tests a second resort that grows by 25% a year from the same start.

$$a_n = 20\,000\,(1.2)^{\,n-1} \qquad R_n = 1500\,a_n \text{ SAR}$$

Year (n)	1	2	3	4
Visitors				

YOUR TASK 5 steps · 20 minutes

1 — FILL THE TABLE (4 min)
Complete the visitors for years 1 to 4. Write the recursive rule and check year 4 with the formula.

2 — FIND THE YEAR (5 min)
Divide by 20 000, write the year as a logarithm, and use the change of base formula. Show the exact expression first.

3 — CHECK AND PRICE IT (4 min)
Test the year before and the year itself. Then find the model revenue in that year in millions of SAR.

4 — WHAT IF? (4 min)
Repeat steps 2 and 3 for 25% growth. Which resort reaches the target first, and by how many years?

5 — PRESENT (3 min)
One sentence for the planners: why does a bigger growth rate reach the target sooner, in terms of the logarithm?

YOUR WORKING write on this sheet

Step 1 — the table and the recursive rule:

Step 2 — the exact expression, then the decimal:

Step 3 — the check and the revenue:

Steps 4 and 5 — the 25% resort and our sentence:

ROLES
- Reader — states the model and the target
- Calculator — fills the table and the decimals
- Checker — tests the year before and the year itself
- Presenter — says the sentence

DONE WHEN ...
- The exact expression is written before the decimal.
- The year is a whole number, justified by testing two consecutive years.
- Units (visitors, SAR, years) appear in every answer.
- Your sentence makes sense to someone who does not study maths.